

Objective: Setting up CI/CD pipelines using GitHub Actions or Jenkins, Automated testing and deployment, Integrating with Docker.

Required Software / Tools: Docker Desktop, Python, VSCode, GITHUB.

Procedure:

Step1: Login to VSCode, open new folder name Exp5 and create app.py file with basic python program as below.

```
def is_palindrome(word: str) -> bool:
    cleaned = word.lower().replace(" ", "")
    return cleaned == cleaned[::-1]

if __name__ == "__main__":
    user_input = input("Enter a word: ")
    if is_palindrome(user_input):
        print(f' '{user_input}' is a palindrome!")
    else:
        print(f' '{user_input}' is not a palindrome.")
```

Step2: Test the python code locally

```
python app.py
```

Step3: create a test code with name test_app.py

```
import app

def test_palindrome_true():
    assert app.is_palindrome("madam") is True
    assert app.is_palindrome("racecar") is True

def test_palindrome_false():
    assert app.is_palindrome("hello") is False
    assert app.is_palindrome("python") is False
```

step4: Test pytest locally

```
pip install pytest
python -m pytest
```

step5: Create requirements.txt with below contents

```
pytest
```

Step6: Create Docker file with below contents name it Dockerfile

```
FROM python:3.10-slim

WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt

COPY . .
CMD ["python", "app.py"]
```

Step7: Build and Test the Docker Container

```
docker build -t palindrome-app .
```

```
docker run -it palindrome-app
```

Step 8: Create CI/CD pipeline using GITHUB Actions with name ci.yml under

i. From terminal run below command

```
mkdir -p .github/workflows
```

ii. Navigate to workflow directory and create ci.yml with below contents

```
name: CI/CD Pipeline
```

```
on:
```

```
  push:
```

```
    branches: [ "main" ]
```

```
  pull_request:
```

```
    branches: [ "main" ]
```

```
jobs:
```

```
  build-test:
```

```
    runs-on: ubuntu-latest
```

```
    steps:
```

```
      - name: Checkout code
```

```
        uses: actions/checkout@v3
```

```
      - name: Set up Python
```

```
        uses: actions/setup-python@v4
```

```
        with:
```

```
          python-version: "3.10"
```

- name: Install dependencies
- run: |
pip install -r requirements.txt

- name: Run tests
- run: python -m pytest

docker-build:

runs-on: ubuntu-latest

needs: build-test

steps:

- name: Checkout code
- uses: actions/checkout@v3

- name: Build Docker image
- run: docker build -t palindrome-app .

Step9: Create Github repo with name Exp5

Step10: Push the code to Github and verify Github Actions

```
echo "# Exp5" >> README.md
```

```
git init
```

```
git add README.md
```

```
git commit -m "first commit"
```

```
git branch -M main
```

```
git remote add origin https://github.com/bandarichandra510/Exp5.git
```

```
git push -u origin main
```

Note: Check if you are able to view readme file in Github UI.

```
git add .
```

```
git commit -m "Initial commit - palindrome app with CI/CD"
```

```
git branch -M main
```

```
git push -u origin main
```